

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Diethyl cyanophosphonate**  
 Product Description: **Diethyl cyanophosphonate**  
 Cat No. : 224290000; 224290050; 224290250  
 Synonyms Diethylphosphoryl cyanide  
 CAS No 2942-58-7  
 Molecular Formula C5 H10 N O3 P

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

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**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                    |                     |
|------------------------------------|---------------------|
| Acute oral toxicity                | Category 2 (H300)   |
| Acute dermal toxicity              | Category 2 (H310)   |
| Acute Inhalation Toxicity - Vapors | Category 2 (H330)   |
| Skin Corrosion/Irritation          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation  | Category 1 (H318)   |

**Label Elements**


Signal Word

Danger

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## Hazard Statements

H314 - Causes severe skin burns and eye damage

H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P262 - Do not get in eyes, on skin, or on clothing

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P284 - Wear respiratory protection

### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

EUH032 - Contact with acids liberates very toxic gas

Combustible liquid

Stench

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                | CAS No    | Weight % |
|--------------------------|-----------|----------|
| Diethyl cyanidophosphate | 2942-58-7 | <=100    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

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|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

## **Most important symptoms and effects, both acute and delayed**

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

## **Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **Extinguishing media**

#### **Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Do not use a solid water stream as it may scatter and spread fire. Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Combustible material. Containers may explode when heated.

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Oxides of phosphorus, Hydrogen cyanide (hydrocyanic acid).

### **Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### **Environmental precautions**

Should not be released into the environment.

### **Methods and Material for Containment and Cleaning Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.

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## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

### Conditions for Safe Storage, Including any Incompatibilities

Keep away from heat, sparks and flame. Keep locked up. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. To maintain product quality: Keep refrigerated.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                | Malaysia | ACGIH TLV | OSHA PEL                           |
|--------------------------|----------|-----------|------------------------------------|
| Diethyl cyanidophosphate |          |           | (Vacated) TWA: 5 mg/m <sup>3</sup> |

| Component                | European Union | The United Kingdom                                                         | Germany                                                                             |
|--------------------------|----------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Diethyl cyanidophosphate |                | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 2 mg/m <sup>3</sup><br>Haut |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

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**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                              |                               |                                          |
|------------------------------|-------------------------------|------------------------------------------|
| Appearance                   | Yellow                        |                                          |
| Physical State               | Liquid                        |                                          |
| Odor                         | Stench                        |                                          |
| Odor Threshold               | No data available             |                                          |
| pH                           | No information available      |                                          |
| Melting Point/Range          | No data available             |                                          |
| Softening Point              | No data available             |                                          |
| Boiling Point/Range          | 104 - 105 °C / 219.2 - 221 °F | @ 19 mmHg                                |
| Flash Point                  | 80 °C / 176 °F                | <b>Method -</b> No information available |
| Evaporation Rate             | No data available             |                                          |
| Flammability (solid,gas)     | Not applicable                | Liquid                                   |
| Explosion Limits             | No data available             |                                          |
| Vapor Pressure               | No data available             |                                          |
| Vapor Density                | No data available             | (Air = 1.0)                              |
| Specific Gravity / Density   | 1.070                         |                                          |
| Bulk Density                 | Not applicable                | Liquid                                   |
| Water Solubility             | Decomposes                    |                                          |
| Solubility in other solvents | No information available      |                                          |

### Partition Coefficient (n-octanol/water)

|                           |                          |                                        |
|---------------------------|--------------------------|----------------------------------------|
| Autoignition Temperature  | No data available        |                                        |
| Decomposition Temperature | No data available        |                                        |
| Viscosity                 | No data available        |                                        |
| Explosive Properties      |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties      | No information available |                                        |

|                   |               |
|-------------------|---------------|
| Molecular Formula | C5 H10 N O3 P |
| Molecular Weight  | 163.11        |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

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Yes. Contact with acids liberates very toxic gas.

## Chemical Stability

Stable under normal conditions. Moisture sensitive.

## Possibility of Hazardous Reactions

|                                 |                               |
|---------------------------------|-------------------------------|
| <b>Hazardous Polymerization</b> | No information available.     |
| <b>Hazardous Reactions</b>      | None under normal processing. |

## Conditions to Avoid

Incompatible products. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

Strong oxidizing agents. Strong acids. Strong bases.

## Hazardous Decomposition Products

Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Oxides of phosphorus. Hydrogen cyanide (hydrocyanic acid).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

|                                        |                                                           |
|----------------------------------------|-----------------------------------------------------------|
| (a) acute toxicity;                    |                                                           |
| Oral                                   | Category 2                                                |
| Dermal                                 | Category 2                                                |
| Inhalation                             | Category 2                                                |
| (b) skin corrosion/irritation;         | Category 1 B                                              |
| (c) serious eye damage/irritation;     | Category 1                                                |
| (d) respiratory or skin sensitization; |                                                           |
| Respiratory                            | No data available                                         |
| Skin                                   | No data available                                         |
| (e) germ cell mutagenicity;            | No data available                                         |
| (f) carcinogenicity;                   | No data available                                         |
|                                        | There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;             | No data available                                         |

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|                                            |                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (h) STOT-single exposure;                  | No data available                                                                                                                                                                                                                                                                                                                                |
| (i) STOT-repeated exposure;                | No data available                                                                                                                                                                                                                                                                                                                                |
| Target Organs                              | No information available.                                                                                                                                                                                                                                                                                                                        |
| (j) aspiration hazard;                     | No data available                                                                                                                                                                                                                                                                                                                                |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                                                                                   |
| Symptoms / effects, both acute and delayed | Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. |
| Endocrine Disrupting Properties            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.                                                                                                                                                                                                              |

## SECTION 12: ECOLOGICAL INFORMATION

|                                        |                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------|
| <u>Ecotoxicity effects</u>             | Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available. |
| <u>Persistence and degradability</u>   | No information available                                                                           |
| Persistence                            | Persistence is unlikely, based on information available.                                           |
| Degradability                          | Decomposes in contact with water.                                                                  |
| Degradation in sewage treatment plant  | No information available. Decomposes in contact with water.                                        |
| <u>Bioaccumulative potential</u>       | Product does not bioaccumulate due to reaction with water                                          |
| <u>Mobility in soil</u>                | Decomposes in contact with water. Is not likely mobile in the environment.                         |
| <u>Endocrine Disruptor Information</u> | This product does not contain any known or suspected endocrine disruptors                          |
| <u>Other adverse effects</u>           | No information available                                                                           |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Waste treatment methods</u>      |                                                                                                                                                                                                           |
| Waste from Residues/Unused Products | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations                                       |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                 |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer Large amounts will affect pH and harm aquatic organisms |

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## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN2922  
Hazard Class 8  
Subsidiary Hazard Class 6.1  
Packing Group II  
Proper Shipping Name Corrosive liquid, toxic, n.o.s. Diethyl cyanophosphonate

### Road and Rail Transport

UN-No UN2922  
Hazard Class 8  
Subsidiary Hazard Class 6.1  
Packing Group II  
Proper Shipping Name Corrosive liquid, toxic, n.o.s. Diethyl cyanophosphonate

### IATA

UN-No UN2922  
Hazard Class 8  
Subsidiary Hazard Class 6.1  
Packing Group II  
Proper Shipping Name Corrosive liquid, toxic, n.o.s. Diethyl cyanophosphonate

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|--------------------------|-----------|------|-----|-------|------|------|-------|------|------|
| Diethyl cyanidophosphate | 220-936-5 | -    | -   | -     | X    | X    | -     | -    | -    |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances



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KECL - Korean Existing and Evaluated Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

22-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**